

# Why Matrix SYK Models Do Not Have R-Charge Concentration

**Abstract.** R-charge concentration — all BPS states of an irreducible complex sitting in a single R-charge — is the proposed fingerprint of a fortuitous, black-hole-like BPS sector, and Chen has conjectured that purely fermionic  $U(N)$  matrix quantum mechanics with a cubic supercharge and three flavours realises it. We compute the BPS cohomology of these models exactly. The  $U(N)$  weight decomposition splits the  $Q$ -complex into pieces of dimension  $\leq 10^4$  in place of  $2^{pN^2}$ , which puts  $N \leq 5$  within reach of exact integer linear algebra; Kostant inversion converts weight-space cohomology into per-irrep BPS counts. The results are uniformly negative and follow a simple law. The BPS states occupy a window of *consecutive* R-charges whose width grows linearly with the gauge rank:  $W = N + 1$  for  $p = 1, 3$ ,  $W = 2N + 1$  for  $p = 2, 4$ ,  $W = 3N + 1$  for  $p = 5$ , with period four in  $p$ . Since the complexes are graded by  $N_\Psi \bmod q$  with  $q = 3$  fixed, a window of width  $W > q$  cannot fit one degree per class, and concentration fails for every  $N \geq 3$ . We then solve the problem analytically: at the maximal weight the complex factorises over the rank  $G$  Cartan directions into  $(\Lambda^\bullet \mathbb{C}^p, \omega \wedge)^{\otimes \text{rank } G}$ , giving  $W = \text{rank } G (w_s - 1) + 1$  in closed form, and an Euler-characteristic argument forces  $w_s \geq 2$ . Concentration therefore requires  $\text{rank } G \leq q - 1$ , and solving for  $w_s$  sharpens this to  $\text{rank } G \leq 2$  — exactly  $u(2)$  and  $su(3)$ . Chen’s supporting  $N = 2$  evidence is one of the only two cases in the entire family where concentration holds, and it holds for kinematic reasons.

## 1 Introduction

**The models.** The degrees of freedom are complex fermionic  $N \times N$  matrices  $\Psi_{ij}^a$ ,  $a = 1, \dots, p$ , with  $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}$  and  $U(N)$  acting by conjugation. All dynamics comes from a nilpotent supercharge,

$$Q = \sum_{a_1 \dots a_q} C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = qQ, \quad (1)$$

with  $N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a$  the R-charge. Only odd  $q$  is admissible:  $\text{Tr}[\Psi^q]$  picks up  $(-1)^{q-1}$  under a cyclic shift, so it vanishes identically for even  $q$ , and  $Q^2 = 0$  holds automatically precisely when  $q$  is odd, because exchanging the two  $q$ -tuples in  $C \otimes C$  is a permutation of  $2q$  fermions with  $q^2$  transpositions, contracted against a totally antisymmetric product. The case  $q = 3, p = 1$  is Chen’s exactly solvable single-matrix model [1], where  $H$  is a function of the gauge Casimir;  $q = 3, p = 3$  with Chen’s couplings is the proposed “matrix SYK” model, for which the Casimir structure is lost.

BPS states are  $E = 0$  states, equivalently classes in  $\ker Q / \text{im } Q$ . Because  $Q$  is a gauge singlet and raises  $N_\Psi$  by  $q$ , the cohomology splits into irreducible complexes labelled by the  $U(N)$  irrep  $\lambda$ , the  $\mathbb{Z}_p$  flavour charge  $\omega$ , and the residue  $c = N_\Psi \bmod q$ .

**Concentration, and what is at stake.** Chang, Chen, Sia and Yang [2] propose *R-charge concentration* as the signature of a fortuitous BPS sector: along each irreducible complex with macroscopic index, all BPS states occupy a *single* degree. In  $\mathcal{N} = 2$  SYK this holds, the BPS counts follow the  $\cos(\pi k/3)$  profile of the  $\mathcal{N} = 2$  super-Schwarzian [5, 4], and the index counts the BPS states exactly. Chen observes that his single-matrix model fails the test — its BPS states span  $N + 1$  consecutive charges [1, (2.22)] — and conjectures that with three flavours “the magic number might be three”, citing an unpublished  $N = 2$  simulation that “suggests” concentration. Whether this survives  $N \geq 3$  has been open, and it is the question we settle.

### Summary of results.

- The BPS window is a run of consecutive R-charges centred on half filling, of width  $W$  growing linearly in  $N$ :  $W = N + 1$  ( $p = 1, 3$ ),  $2N + 1$  ( $p = 2, 4$ ),  $3N + 1$  ( $p = 5$ ), with period four in  $p$  (Table 1). Exact counts at  $N = 2, 3, 4, 5$  (Table 2).
- Since the grading period is  $q = 3$ ,  $W > 3$  makes one-degree-per-class impossible. Concentration holds at  $N = 2$  and fails for every  $N \geq 3$  — by counting, not by dynamics.
- Going traceless buys exactly one unit:  $su(N)$  has window  $N$ , so  $su(3)$  concentrates and  $su(4)$  does not. The controlling parameter is  $\text{rank } G$ , not  $N$ .
- Analytically, the maximal-weight complex factorises over Cartan directions,  $Z_{\max}(t) = t^{k_0} (z_{\text{slot}}(t))^{\text{rank } G}$ , giving  $W = \text{rank } G (w_s - 1) + 1$ ; an Euler argument gives  $w_s \geq 2$ , hence  $\text{rank } G \leq q - 1$ , and solving for  $w_s$  gives  $\text{rank } G \leq 2$ .

| $p$              | 1     | 2      | 3     | 4      | 5      | 6      | 7     | 8      |
|------------------|-------|--------|-------|--------|--------|--------|-------|--------|
| $w_s$            | 2     | 3      | 2     | 3      | 4      | 3      | 2     | 3      |
| window width $W$ | $N+1$ | $2N+1$ | $N+1$ | $2N+1$ | $3N+1$ | $2N+1$ | $N+1$ | $2N+1$ |

**Table 1:** BPS window width for  $q = 3$  as a function of the number of flavours, from the closed form  $W = \text{rank } G(w_s - 1) + 1$  of Sec. 4 with  $\text{rank } U(N) = N$ . The pattern has period four in  $p$ :  $W = N + 1$  for  $p \equiv 3 \pmod{4}$  and  $p = 1, 2N + 1$  for even  $p$ ,  $3N + 1$  for  $p \equiv 1 \pmod{4}$ ,  $p > 1$ . Verified directly at  $N = 3$  for  $p = 1, 2, 3, 4$ . In every case  $W > q = 3$  once  $N \geq 3$ .

## 2 Methodology

**Weight-space decomposition.** Direct computation of  $\ker Q / \text{im } Q$  is hopeless: at  $N = 3$ ,  $p = 3$  the middle sector has  $\dim \mathcal{H}_{13} = 2 \times 10^7$ , and at  $N = 4$  it is  $3 \times 10^{13}$ . The observation that removes the problem is that  $Q$  in (1) is a  $U(N)$  singlet — every term is a closed index loop — so it commutes with the Cartan and preserves the  $U(N)$  weight. The complex therefore splits over weights,

$$\cdots \xrightarrow{Q} W_\lambda(k) \xrightarrow{Q} W_\lambda(k+q) \xrightarrow{Q} \cdots, \quad \dim H^k(W_\lambda) = \dim W_\lambda(k) - \text{rank } Q_k - \text{rank } Q_{k-q}, \quad (2)$$

and the weight spaces are small: the largest relevant one is 126 at  $N = 3$  and  $3 \times 10^4$  at  $N = 4$ . Since  $\Psi_{ij}^a$  carries weight  $e_i - e_j$ , a state's weight depends only on the occupation matrix  $n_{ij}$ , and enumerating  $W_\lambda(k)$  is a pruned search over  $n_{ij}$  rather than a scan of Fock space.

**Exact ranks.** Ranks are computed by Gaussian elimination over  $\mathbb{F}_P$ . A rank over  $\mathbb{F}_P$  can only under-estimate the rational rank, so every value reported here was recomputed over a second prime and, where feasible, in floating point. For the larger weight spaces we use a right-looking blocked LU whose trailing update is a single BLAS matmul; exactness is preserved by choosing  $P \approx 2^{20}$  so that  $\text{block} \times (P - 1)^2 < 2^{53}$  and every float64 partial sum is an exact integer. A further factor of  $p^2$  in time and memory comes from the  $\mathbb{Z}_p$  flavour symmetry:  $Q$  commutes with the flavour rotation, so each weight space splits into  $p$  blocks of size  $\approx n/p$ , written exactly over  $\mathbb{F}_P$  with  $P \equiv 1 \pmod{p}$  so that a primitive  $p$ -th root of unity exists. Together these take the largest tractable weight space from 924 to 30 294; the complete  $N = 4$  run below takes 30 minutes at 6 GB.

**From weights to irreps.** A weight space collects contributions from every irrep containing that weight,

$$\dim H^k(W_\lambda) = \sum_{\mu \geq \lambda} K(\mu, \lambda) h_\mu^k, \quad K(\mu, \lambda) = \sum_{w \in W} \text{sgn}(w) \mathcal{P}(w(\mu + \rho) - (\lambda + \rho)), \quad (3)$$

with  $K$  the Kostant weight multiplicity and  $\mathcal{P}$  the Kostant partition function. The system is triangular and is inverted from the top down. Crucially it is not a truncation: higher weights have *smaller* weight spaces, so the affordable set is automatically closed upward.

**Validation.** Every number below passes four independent checks: (i) the weight-space dimensions reproduce the multiplicities obtained from the Fock-space character by a Weyl alternating sum; (ii) for each irrep and each class  $c$ , the Euler characteristic  $\sum_{k \equiv c} (-1)^k h^k$  equals the independently computed refined Witten index, including when refined by flavour charge; (iii) all peeled multiplicities are non-negative, a strong constraint since the peel subtracts large numbers; (iv) the particle-hole symmetry  $h^k = h^{pN^2 - k}$  emerges without being imposed. At  $N = 2$  the method reproduces exact diagonalisation, and for  $p = 1$  it reproduces Chen's closed form [1, (2.21)–(2.22)] at  $N = 2, 3, 4$ .

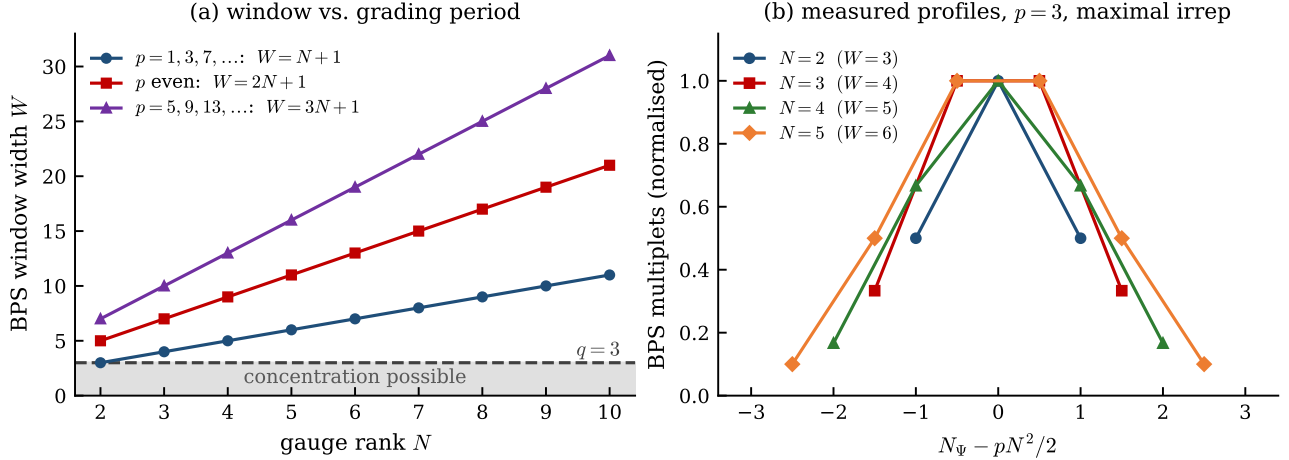
## 3 Results

**The window and its width.** For every model and every irrep we have computed, the BPS cohomology occupies a run of *consecutive* R-charges centred on half filling  $pN^2/2$ . The width obeys a simple law, periodic in  $p$  with period four (Table 1), and grows linearly in  $N$  in all cases. Table 2 gives the exact multiplet counts for the maximal irrep at  $p = 3$ ; the profiles are binomial,  $3^N \binom{N}{j}$ , and the totals are  $6^N$ .

**Why concentration fails, and why  $N = 2$  is misleading.** The complexes are graded by  $c = N_\Psi \pmod{3}$ . A window of  $W$  consecutive charges distributes over the three classes with  $\lceil W/3 \rceil$  charges in the most populated one, so one-degree-per-class requires  $W \leq 3$ . At  $N = 2$ ,  $W = 3$  exactly: each class receives exactly one charge and concentration holds — for purely kinematic reasons, with no input from the dynamics. This is the regime of the  $N = 2$  evidence cited for the matrix-SYK proposal. From  $N = 3$  it is arithmetically impossible. At  $N = 3$  the surplus falls in the  $c = 0$  class, whose index vanishes, so the extra states are invisible to the index and the non-zero-index classes still saturate; at  $N = 4$  the surplus falls in  $c = 1, 2$ , whose index does *not* vanish, so saturation fails where

| $N$ | window ( $N_\Psi$ ) | width | BPS multiplets $h^k$             | profile            |
|-----|---------------------|-------|----------------------------------|--------------------|
| 2   | 5, 6, 7             | 3     | 9, 18, 9                         | $3^2 \binom{2}{j}$ |
| 3   | 12, ..., 15         | 4     | 27, 81, 81, 27                   | $3^3 \binom{3}{j}$ |
| 4   | 22, ..., 26         | 5     | 81, 324, 486, 324, 81            | $3^4 \binom{4}{j}$ |
| 5   | 35, ..., 40         | 6     | 243, 1215, 2430, 2430, 1215, 243 | $3^5 \binom{5}{j}$ |

**Table 2:** Exact BPS multiplet counts for the maximal irrep of the three-flavour model,  $\lambda_i^* = p(N+1-2i)$ . The window is  $N+1$  consecutive charges centred on  $pN^2/2$ , and  $Z_{\text{BPS}} = 3^N(1+t)^N t^{k_0}$ , i.e.  $6^N$  multiplets. The  $(1+t)^N$  factor is identical to Chen’s single-matrix result [1, (2.20)] — the model he identifies as *not* concentrated.



**Figure 1:** (a) BPS window width against gauge rank. The grading period  $q=3$  is fixed (dashed), so concentration is possible only inside the shaded band; every flavour number leaves it at or before  $N=3$ . (b) Measured BPS multiplet profiles for  $p=3$  at the maximal irrep, normalised and centred on half filling: the window visibly widens by one charge per unit of  $N$ .

the index is macroscopic. Across the thirteen highest irreps at  $N=4$  we count 3,008,825,568 BPS states of which 19.0% are invisible to the index.

**Robustness.** The conclusion does not depend on the couplings, the gauge group, or the supercharge degree. (i) *Couplings:* the cohomology depends on  $C$  only through  $\omega = \text{Alt}(C) \in \Lambda^q \mathbb{C}^p$ , and only through its  $GL(p)$  orbit. At  $p=3$ ,  $\dim \Lambda^3 \mathbb{C}^3 = 1$ , so  $\omega$  is unique up to scale: Chen’s tensor is the *only* non-degenerate three-flavour cubic supercharge, and randomising  $C$  either reproduces it or sets  $\omega=0$  and widens the window to  $3N+1$  (sampling 30 random cyclic tensors: 26 and 4 respectively). More generally random couplings are *optimal* — ranks are lower semicontinuous in  $\omega$ , so specialising can only enlarge the cohomology and widen the support — so every width quoted here is the best case. Disorder, the ingredient that makes SYK work, has nowhere to act. (ii) *Casimir:* descending the spectrum at  $N=3$  through complete Casimir tiers from  $C_2=48$  to  $C_2=24$  — thirteen irreps, up to Fock multiplicity 110 080, a factor 215 above the maximal irrep — the support is width 4 in every single case. (iii) *Gauge group:*  $su(N)$  removes one Cartan direction and narrows the window by exactly one, so  $su(3)$  (rank 2) concentrates with  $h=9, 18, 9$  while  $su(4)$  (rank 3) does not; the controlling parameter is rank  $G$ . (iv) *Degree:* raising  $q$  does not help, for the reason given next.

## 4 The analytic no-go

**Factorisation over Cartan slots.** At the maximal weight  $\lambda_i^* = p(N+1-2i)$  every off-diagonal mode is saturated, so the only unoccupied modes are the diagonal ones. A gauge-invariant  $Q$  must create a closed index loop, and any loop entering an index cannot leave it — its outgoing edges are occupied. The surviving loops are therefore *self-loops* at a single index, creating  $q$  diagonal modes of distinct flavours. Hence  $Q = \sum_{i=1}^{\text{rank } G} Q_i$  with  $Q_i$  acting only within slot  $i$ , and  $Q_i$  is wedging with the totally antisymmetric part of  $C$ . By Künneth,

$$Z_{\text{max}}(t) = t^{k_0} (z_{\text{slot}}(t))^{\text{rank } G}, \quad z_{\text{slot}}(t) = \sum_j \dim H^j(\Lambda^\bullet \mathbb{C}^p, \omega \wedge) t^j, \quad \omega = \text{Alt}(C) \in \Lambda^q \mathbb{C}^p. \quad (4)$$

Equation (4) reproduces all ten measured  $u(N)$  cases exactly, across  $p=1, 2, 3, 4$ ,  $q=3, 5$  and  $N=2, \dots, 5$ . If  $z_{\text{slot}}$  has support on  $w_s$  degrees then the window width is  $W = \text{rank } G (w_s - 1) + 1$ , which is the law in Table 1.

$w_s \geq 2$ , and the bound. Let  $E_c = \sum_{j=c(q)} (-1)^j \binom{p}{j}$  be the Euler characteristic of the class- $c$  subcomplex of  $\Lambda^\bullet \mathbb{C}^p$ ;  $E_c \neq 0$  forces cohomology somewhere in class  $c$ . Two elementary facts pin this down:  $\sum_c E_c = (1-1)^p = 0$  for  $p \geq 1$ , so exactly one non-zero  $E_c$  is impossible; and  $E_c = \frac{1}{q} \sum_{m=1}^{q-1} \zeta^{-mc} (1-\zeta^m)^p$  with  $\zeta = e^{2\pi i/q}$  and  $1-\zeta^m \neq 0$ , so they cannot all vanish. At least two classes therefore carry cohomology,  $w_s \geq 2$ , and

$$W \geq \text{rank } G + 1 \quad \implies \quad \text{concentration} \Rightarrow \text{rank } G \leq q - 1. \quad (5)$$

Raising  $q$  cannot rescue this. With  $\omega \neq 0$  (which requires  $q \leq p$ ) the minimum of  $w_s$  is  $q-1$  for  $q \geq 5$ , so  $\text{rank } G \leq \lfloor (q-1)/(q-2) \rfloor = 1$ ; only  $q=3$  attains  $w_s=2$ , and there  $\text{rank } G \leq 2$ . Scanning all  $q \leq p \leq 14$ , the maximum rank compatible with concentration is **2**, attained at  $q=3$  with  $p \equiv 3 \pmod{4}$ . Scaling  $p$  yields infinitely many concentrating models, all at rank 2.

**The free branch.** One evasion exists and is instructive. If  $q > p$  then  $\omega \in \Lambda^q \mathbb{C}^p = 0$ , the differential vanishes on the sector,  $z_{\text{slot}} = (1+t)^p$ , and for  $p=1$  this gives  $w_s=2$  with arbitrary  $q$ , hence unbounded rank. But  $\omega=0$  means  $Q$  annihilates the sector outright: every state is BPS because nothing acts on it, not because of fortuity. This is the single-matrix, Casimir-solvable situation. It satisfies the counting definition of concentration while being physically its opposite, which shows that the counting criterion alone is not sufficient — one needs  $Q$  to act non-trivially *and* the window to fit inside one period, and (5) says those demands are incompatible beyond rank 2.

## 5 Outlook

- **Close the last gap in the theorem.** The bound  $\text{rank } G \leq q-1$  is proved; the sharpening to  $\text{rank } G \leq 2$  uses the observed  $w_s^{\min} = q-1$  for  $q \geq 5$ , verified at  $q=3, 5, 7, 9$  and by an exhaustive scan of  $q \leq p \leq 14$ . Proving  $w_s^{\min} = q-1$  is a self-contained question about  $H(\Lambda^\bullet \mathbb{C}^p, \omega \wedge)$  with no physics in it.
- **Beyond the maximal weight.** The factorisation (4) is derived at the maximal weight for  $u(N)$ ; it fails for  $su(N)$ , where Cartan directions straddle slots — visible in the anomalous  $su(4)$  profile 13, 81, 81, 13. Empirically the window matches the maximal one throughout the  $N=3$  spectrum down to  $C_2=24$ , but this is not derived.
- **The low-Casimir bulk.** The macroscopic-index complexes lie at low Casimir — at  $N=4$  the largest multiplicity is  $6.6 \times 10^9$  at  $C_2=20$  — where weight spaces reach  $10^9$  and a genuinely sparse rank routine is needed.
- **Chaos without concentration.** The three-flavour  $H$  is *not* a Casimir function, so the model may still be chaotic and have a super-Schwarzian near-BPS spectrum even without a concentrated BPS sector. Nothing here bears on that; Altland–Zirnbauer statistics of the singular values of  $Q_k$  are now cheap to extract weight by weight.
- **What would concentrate.** Equation (5) says one needs  $q$  growing with the rank, and  $\omega \neq 0$  needs  $p \geq q$ : an interaction touching  $O(N)$  of the  $N^2$  matrix entries. Whether such a model is chaotic, or has a sensible large- $N$  limit, is open.

## 6 Numerical implementation

All computations are exact integer linear algebra; no floating-point tolerance enters any reported number. Weight bases are enumerated by a pruned recursion over the occupation matrix, validated by  $\sum_\lambda \dim W_\lambda(k) = \binom{pN^2}{k}$  for every  $k$  at  $N=3$  and against the Fock character at  $N=4$ . Ranks use Gaussian elimination over  $\mathbb{F}_P$  with  $P=2^{31}-1$  for small blocks, and for large ones a blocked LU with the trailing update as a BLAS matmul over  $P \approx 2^{20}$ , chosen so that  $\text{block} \times (P-1)^2 < 2^{53}$  keeps every float64 partial sum an exact integer; every rank is confirmed over a second prime. The  $\mathbb{Z}_p$  flavour blocking is performed over  $\mathbb{F}_P$  with  $P \equiv 1 \pmod{p}$ , the eigenbasis built from orbits of the rotation with their fermionic signs, and is validated by requiring the block ranks to sum to the dense rank for every  $k$ . Kostant multiplicities are computed from the Weyl alternating sum with a pruned partition function and verified against  $\sum_\lambda K(\mu, \lambda) = \dim \mu$ . Representative costs: the eight highest irreps at  $N=3$  in under two minutes at 1.0 GB; thirteen irreps at  $N=4$  in 30 minutes at 6.1 GB; the  $N=5$  maximal weight in 49 seconds at 0.4 GB.

## References

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